

Lecture Note: IM Session 1 – Intro To Python (Variables & Data Types)

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Notes

Foundation

Recommended

Resource



Associated Lectures (1)



Description



Discussions



Lecture Notes: Introduction to Python, Setup & Syntax



1. Big Picture: Why Python Matters

Python is not just another programming language - it's the **gateway language** to modern technology. Here's why it stands out:

Feature	What It Means	Why It Helps You
Readable	Looks like English	Easy for beginners
Interpreted	Runs line by line	Instant feedback, less waiting
Cross-Platform	Works on any OS	Learn once, use anywhere
Versatile	Used in AI, Web, Data Science, Automation	Opens career options



2. Setting Up Python the Right Way (Windows & macOS)

For Windows Users:

Go to python.org/downloads.

Download the latest **Python 3.x** installer.

Check “Add Python to PATH” during installation 

Open **Command Prompt** and verify:

```
python --version
```

You should see something like `Python 3.12.1`

For macOS Users:

Visit python.org/downloads.

Download the macOS installer and run it.

After installation, open **Terminal** and type:

```
python3 --version
```

You should see your Python version number.

If it doesn't work, install via **Homebrew** (optional):

```
brew install python
```

Set Up VS Code for Python

Download VS Code from code.visualstudio.com.

Open VS Code → Go to **Extensions** → Search "Python" → Install it.

Create a file named `hello.py` and add:

```
print("Hello, Python!")
```

Run it:

- **Option 1:** Click  "Run Python File" at the top-right
- **Option 2:** Open terminal in VS Code and type:

```
python hello.py
```

(Use python3 on Mac if needed)



3. Understanding Python Syntax & Indentation

Python uses **indentation** instead of `{}` or `;` like other languages

Rules You Must Remember

- Every indented block = a “**group**” of code that works together.
- Use **4 spaces** (not tabs) per indentation level.
- A **colon (:)** means “something is coming next” - usually an indented block.

Example:

```
if True:
    print("Indentation defines structure!")
```

Comments in Python

Type	Symbol	Example
Single-line	#	# This is a comment
Multi-line / Docstring	""" ... """	"""This is a multi-line comment"""

Example:

```
# Single line
"""This is a
multi-line comment"""
```



Pro Tip: Use comments to explain *why* you did something, not *what*.



4. Running Python Code - Two Ways

1. Interactive Shell

Best for quick testing and learning.

```
>>> print("Hello")
Hello
>>> 2 + 3
```

2. Script File (.py)

Best for reusable, long programs.

```
# hello.py
print("Hello, Python Learner!")
```

Run it in terminal:

```
python hello.py
```



5. Common Mistakes & Fixes

Mistake	What Happens	Correct Fix
Forgetting parentheses in print	SyntaxError	Use print("Hello")
Mixing tabs and spaces	IndentationError	Use only 4 spaces
Forgetting colon after if, for, etc.	SyntaxError	if True:
Not adding Python to PATH (Windows)	Command not found	Reinstall with PATH checked
Wrong command on macOS	Command not found	Use python3 instead of python



6. Practice Time: Try These on Your Own



Task 1:

Create a script `introduction.py` that:

- Prints your name and favorite color.
- Adds a comment explaining each line.
- Runs without indentation or syntax errors.

Hint:

```
# Displaying my name
print("Name: John")
# Displaying my favorite color
```

```
print("Favorite color: Blue")
```



Task 2:

Try this in the **Python shell**:

```
>>> 10 + 5
>>> 10 - 5
>>> 10 / 5
>>> 10 // 3
>>> 10 ** 2
```

Notice the difference between `/`, `//`, and `**`

✖ 7. Debugging Checklist

- ✓ Use `print()` to check intermediate results
- ✓ Check indentation before running
- ✓ Make sure you're in the correct folder
- ✓ Save your file before execution
- ✓ Re-run the program after changes

If something doesn't work - read the **error message carefully**. It usually tells you the line number and type of error.



8. Python Installation & VS code Setup

◆ For Windows Users

- [Python Installation on Windows](#)
- [VS Code Python Setup](#)

◆ For macOS Users

- [Download & Install Python](#)
- [VS Code Python Setup & Debug on Mac](#)



9. Quick Reference

Command	Purpose	Example
python --version	Check Python version	Python 3.12.1
print()	Output data	print("Hello")
input()	Take user input	name = input("Enter name: ")
#	Comment	# This is a note
""" """	Multi-line comment	"""Explain logic"""
python file.py	Run a Python file	python hello.py



10. Wrap-Up: What You Should Take Away

- Python is **simple yet powerful**.
- Correct setup = smooth learning experience.
- Indentation and comments are **your best friends**.
- Errors are normal - they're your teachers.
- Next step: **Variables & Data Types** - start storing and manipulating information.
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